BLOCKCHAIN Experiment No: 3

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D20A/22

Aim: Create a Cryptocurrency using Python and perform mining in the Blockchain created.

Guidelines

Lab Objectives: To implement public and private Blockchain.

Lab Outcomes (LO): Demonstrate the concept of Blockchain in real-world Applications (LO4)

Task to be performed :

1. Download the code from folder, Lab_3
2. Install requests in the virtual environment created in the Lab 2. (Follow the instructions)
3. Run the files - hadcoin_node_5001.py, hadcoin_node_5002.py, hadcoin_node_5003.py in 3 different terminals.
4. Open Postman, from each node - invoke connect_node() and pass the peers as POST requests.
5. Perform the following functions
    - Add Transactions - invoke add_transactions() as a POST request.
    - mining - mine_block(),
    - fetch the chain - get_chain(),
    - replace the longest chain - replace_chain()
6. Modify the code such that transactions are removed after they are added to the block.

Tools & Libraries used :

- Install Flask : pip install Flask
- Download Postman from https://www.postman.com/
- Python Libraries : datetime, jsonify, hashlib, uuid4, urlparse, request
- Install requests : pip install requests==2.18.4

Instructions : (Prepare for viva for the following topics)

1. Challenges in P2P networks
2. How transactions are performed on the network?
3. Explain the role of mempools
4. Write briefly about the libraries and the tools used during implementation.

Outcome :

1. Understood the challenges in P2P networks, how transactions are performed and how a miner mines a block to be added in a blockchain.
2. Implemented a Cryptocurrency in Python using Flask, Postman and Python libraries such as datetime, jsonify, hashlib, uuid4, urlparse, request.
3. Successfully mined the blocks among a P2P network with 3 nodes.
4. Performed transactions via the network.

5. Successfully updated the block across the network
6. Prepare a document with Aim, Tasks performed, Program, Output and Conclusion.
7. Submit the hardcopy by the 2nd week of August 2023
(As per the instructions, submit a hard copy of the same).

Theory:

1. Blockchain Overview

Blockchain is a **distributed and decentralized ledger** that stores information in a series of linked blocks.
Each block contains:

- Transaction data
- Timestamp
- Previous block's hash
- Its own unique hash (digital fingerprint)

Once data is recorded in a blockchain, it becomes **immutable** because altering one block would require recalculating all subsequent blocks.

2. Mining

Mining is the process of:

1. Collecting pending transactions into a block.
2. Performing a computational puzzle (Proof-of-Work) to find a valid hash.
3. Adding the new block to the blockchain.
   Broadcasting it to all connected peers.

Miners are rewarded with cryptocurrency for successfully mining a block.

3. Multi-Node Blockchain Network

In this lab, we simulate **three independent blockchain nodes** (`5001`, `5002`, `5003`).
Each node:

- Runs on a separate port.
- Maintains its own copy of the blockchain.
- Can connect with peers to share and validate blocks.

4. Consensus Mechanism

We use the **Longest Chain Rule**:

- If multiple versions of the chain exist, the **longest valid chain** is chosen.
- This ensures all nodes agree on a single transaction history.

5. Transactions & Mining Reward

Each transaction has:

- Sender
- Receiver
- Amount

When mining a block:

- Pending transactions are added to the block.
- A **reward transaction** is added automatically to pay the miner.

6. Chain Replacement

When `/replace_chain` is called:

1. Node requests chains from peers.
2. If it finds a longer and valid chain, it replaces its own.
3. This keeps the blockchain consistent across all nodes.

Tools & Libraries Used

- **Python 3.x**
- **Flask** – Web framework for API endpoints
  `pip install Flask`
- **Requests** – For HTTP communication between nodes
  `pip install requests==2.18.4`
- **Postman** – For testing API requests
- Python Standard Libraries:
  - `datetime`
  - `jsonify`
  - `hashlib`
  - `uuid4`
  - `urlparse`
  - `request`

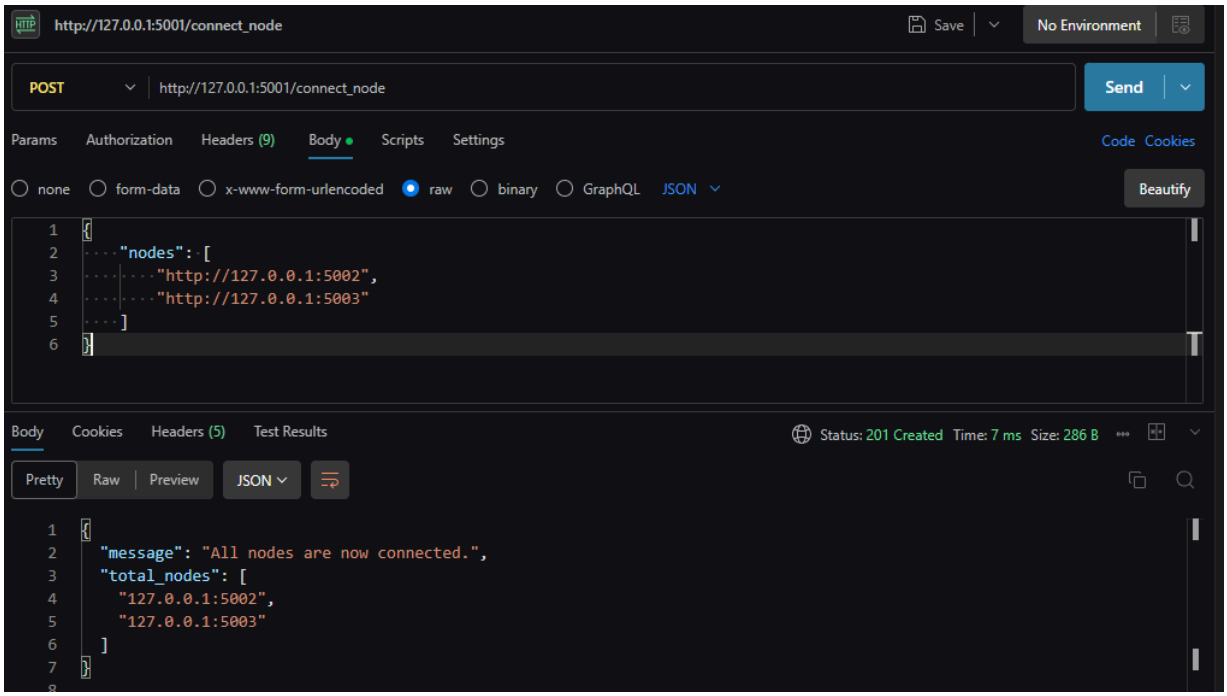
Procedure and Output:

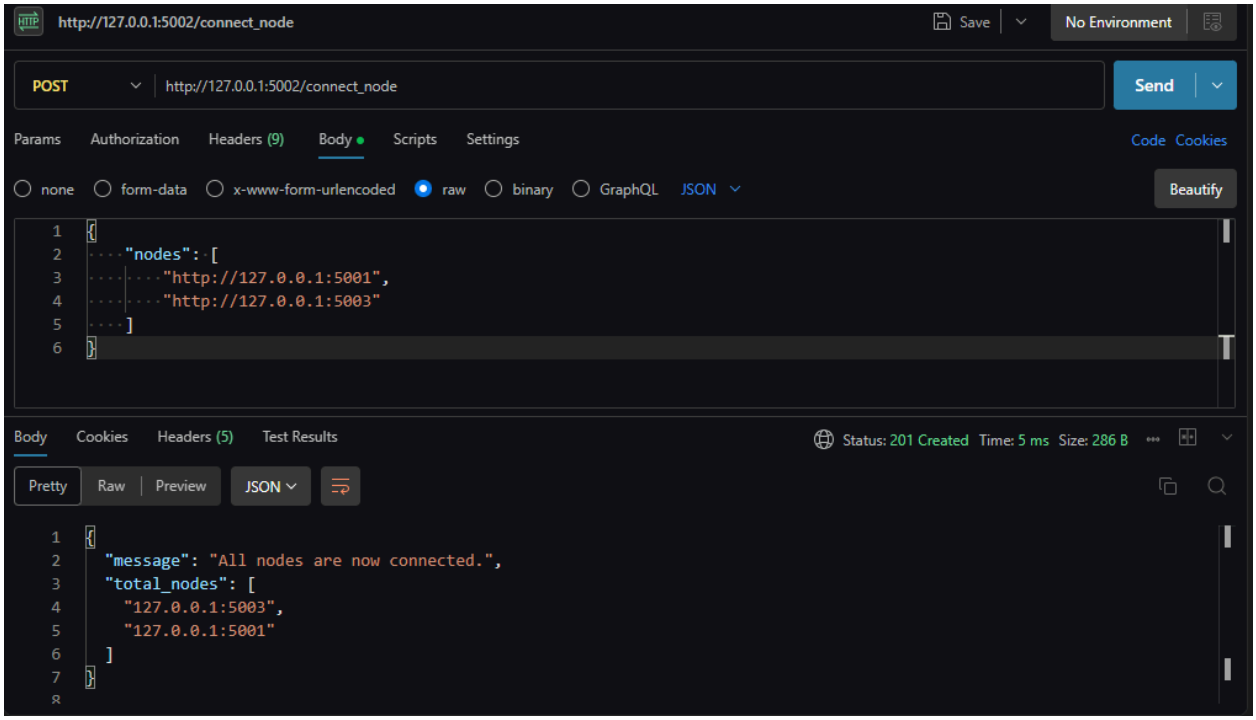
1. Download `hadcoin_node_5001.py`, `hadcoin_node_5002.py`, `hadcoin_node_5003.py`.
2. Install required packages.
3. Run each node in separate terminals.

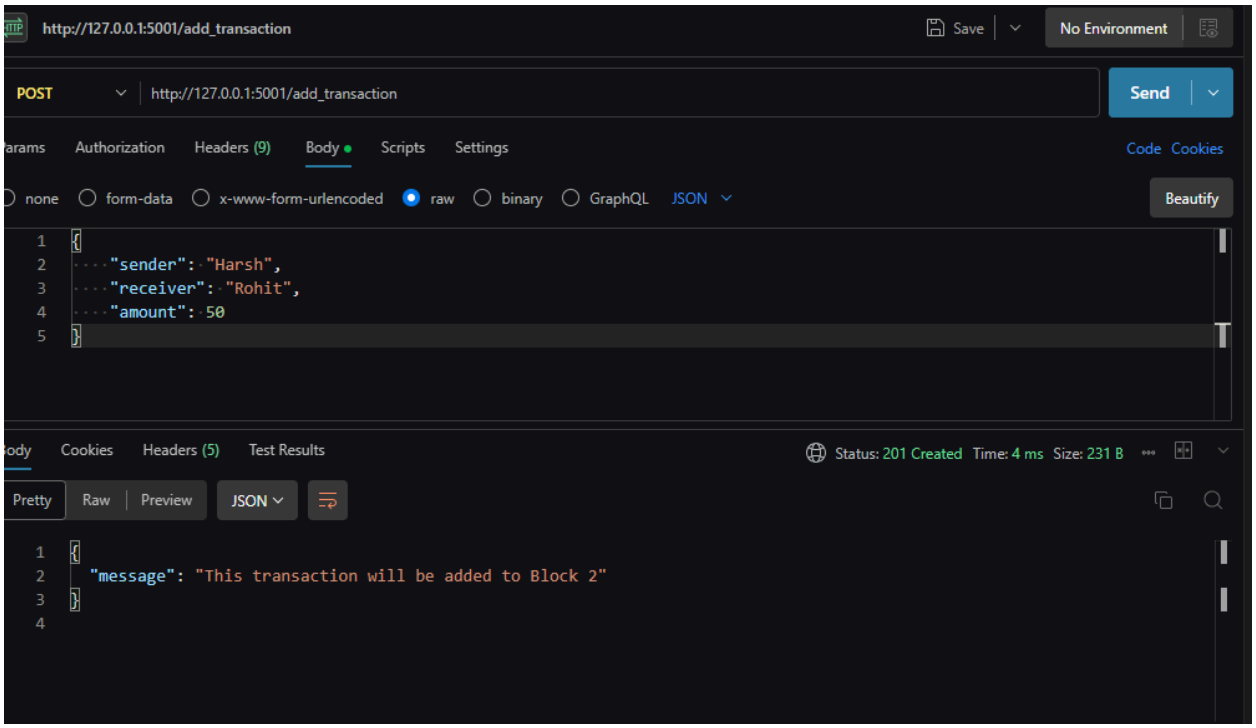
http://127.0.0.1:5001/

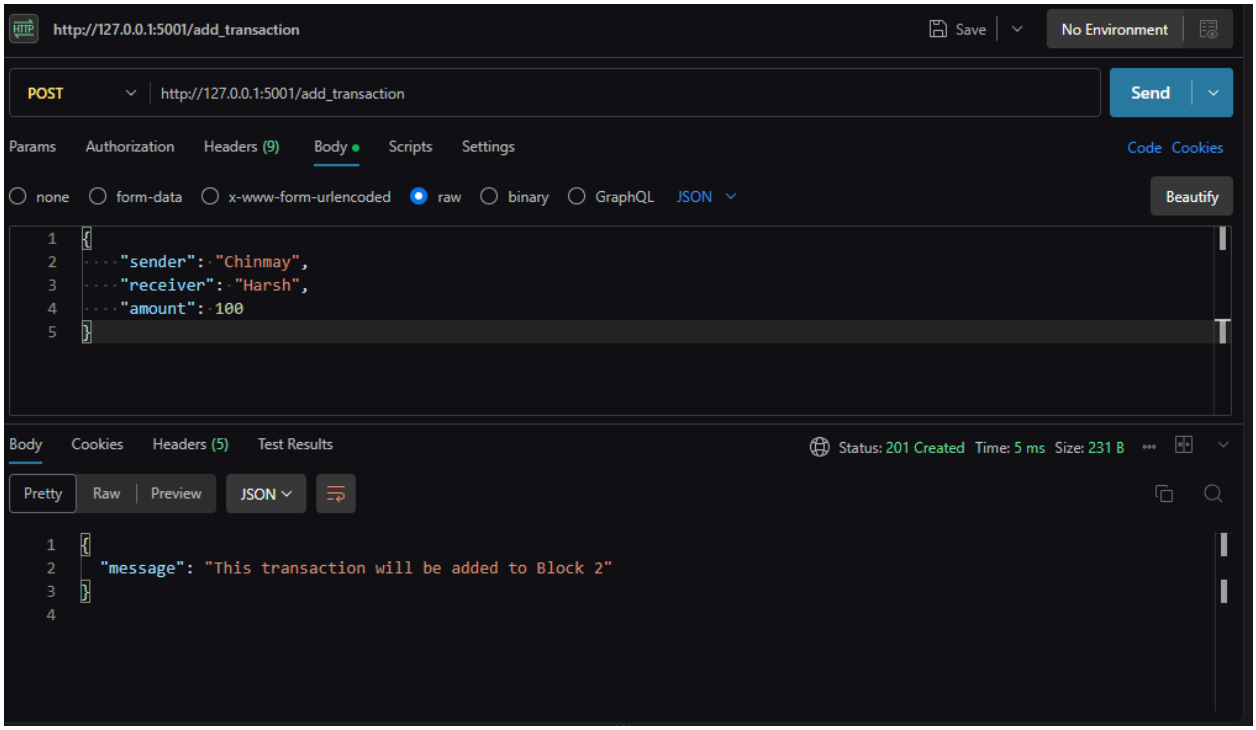
Not Found

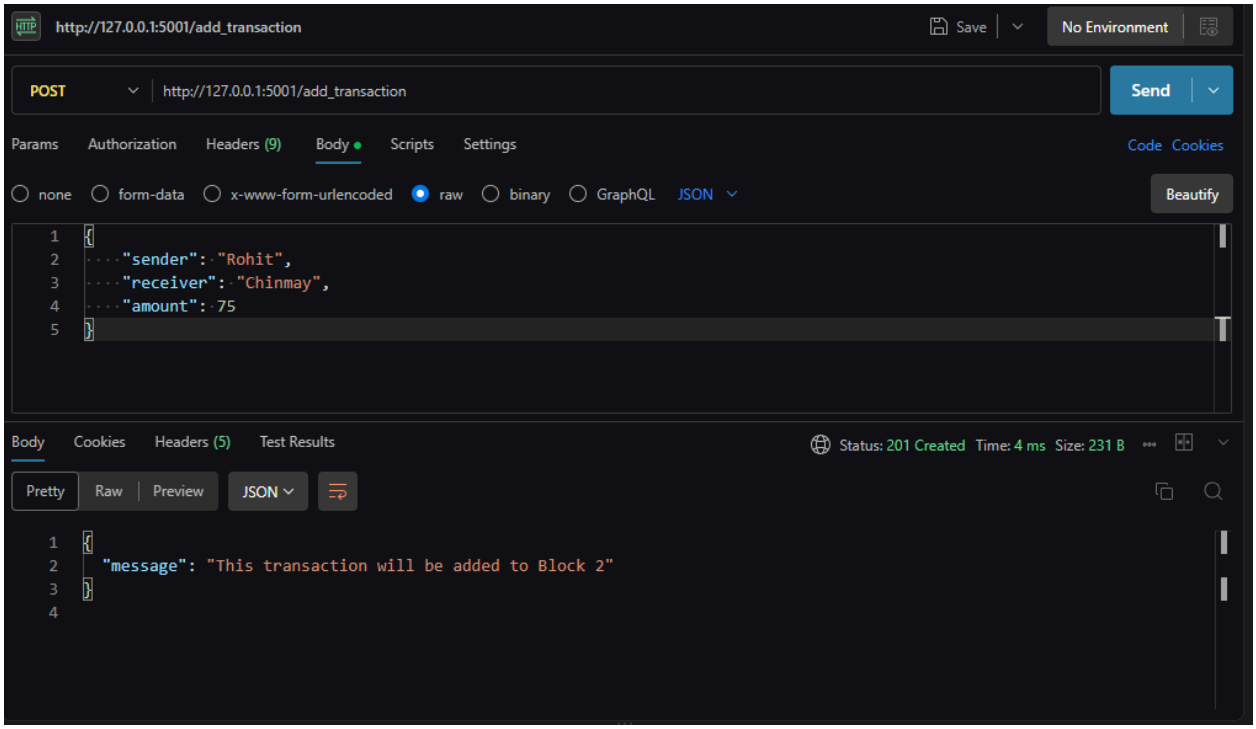
The requested URL was not found on the server. If you entered the URL manually please check your spelling and try again.




http://127.0.0.1:5002/connect_node
Save
No Environment
POST
http://127.0.0.1:5002/connect_node
Send
Params
Authorization
Headers (9)
Body
Scripts
Settings
Code
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Beautify
{
"nodes": [
"http://127.0.0.1:5001",
"http://127.0.0.1:5003"
]
}
Body
Cookies
Headers (5)
Test Results
Status: 201 Created
Time: 5 ms
Size: 286 B
Pretty
Raw
Preview
JSON
{
"message": "All nodes are now connected.",
"total_nodes": [
"127.0.0.1:5003",
"127.0.0.1:5001"
]
}


http://127.0.0.1:5001/add_transaction
Save
No Environment
POST
http://127.0.0.1:5001/add_transaction
Send
Authorization
Headers (9)
Body
Scripts
Settings
Code
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Beautify
{
"sender": "Harsh",
"receiver": "Rohit",
"amount": 50
}
Cookies
Headers (5)
Test Results
Status: 201 Created
Time: 4 ms
Size: 231 B
Pretty
Raw
Preview
JSON
{
"message": "This transaction will be added to Block 2"
}


http://127.0.0.1:5001/add_transaction
Save
No Environment
POST
http://127.0.0.1:5001/add_transaction
Send
Params
Authorization
Headers (9)
Body
Scripts
Settings
Code
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Beautify
{
"sender": "Chinmay",
"receiver": "Harsh",
"amount": 100
}
Body
Cookies
Headers (5)
Test Results
Status: 201 Created
Time: 5 ms
Size: 231 B
Pretty
Raw
Preview
JSON
{
"message": "This transaction will be added to Block 2"
}


http://127.0.0.1:5001/add_transaction
Save
No Environment
POST
http://127.0.0.1:5001/add_transaction
Send
Params
Authorization
Headers (9)
Body
Scripts
Settings
Code
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Beautify
{
"sender": "Rohit",
"receiver": "Chinmay",
"amount": 75
}
Body
Cookies
Headers (5)
Test Results
Status: 201 Created
Time: 4 ms
Size: 231 B
Pretty
Raw
Preview
JSON
{
"message": "This transaction will be added to Block 2"
}

http://127.0.0.1:5001/mine_block

Save | No Environment

GET http://127.0.0.1:5001/mine_block Send

Params Authorization Headers (7) Body Scripts Settings Code Cookies

Body Cookies Headers (5) Test Results Status: 200 OK Time: 6 ms Size: 784 B

Pretty Raw Preview JSON


```
{
  "index": 2,
  "message": "Congratulations, you just mined a block!",
  "previous_hash": "0ed05e2dd6e63bbe68661bd48c835ccde704af3c4d9887e2058610901eeb77be",
  "proof": 533,
  "timestamp": "2026-08-24 12:20:33.119725",
  "transactions": [
    {
      "amount": 50,
      "receiver": "Rohit",
      "sender": "Harsh"
    },
    {
      "amount": 100,
      "receiver": "Harsh",
      "sender": "Chinmay"
    },
    {
      "amount": 75,
      "receiver": "Chinmay",
      "sender": "Rohit"
    },
```


http://127.0.0.1:5001/get_chain

Save | No Environment

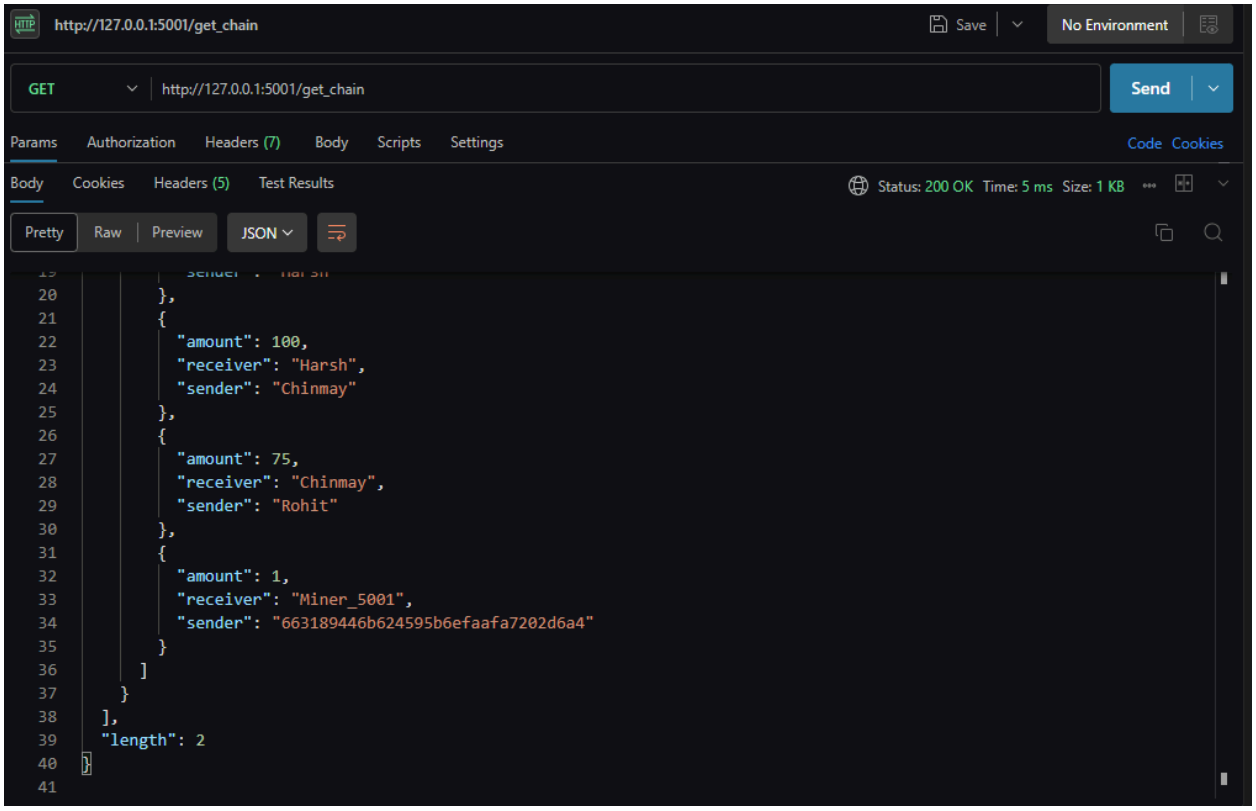
GET http://127.0.0.1:5001/get_chain Send

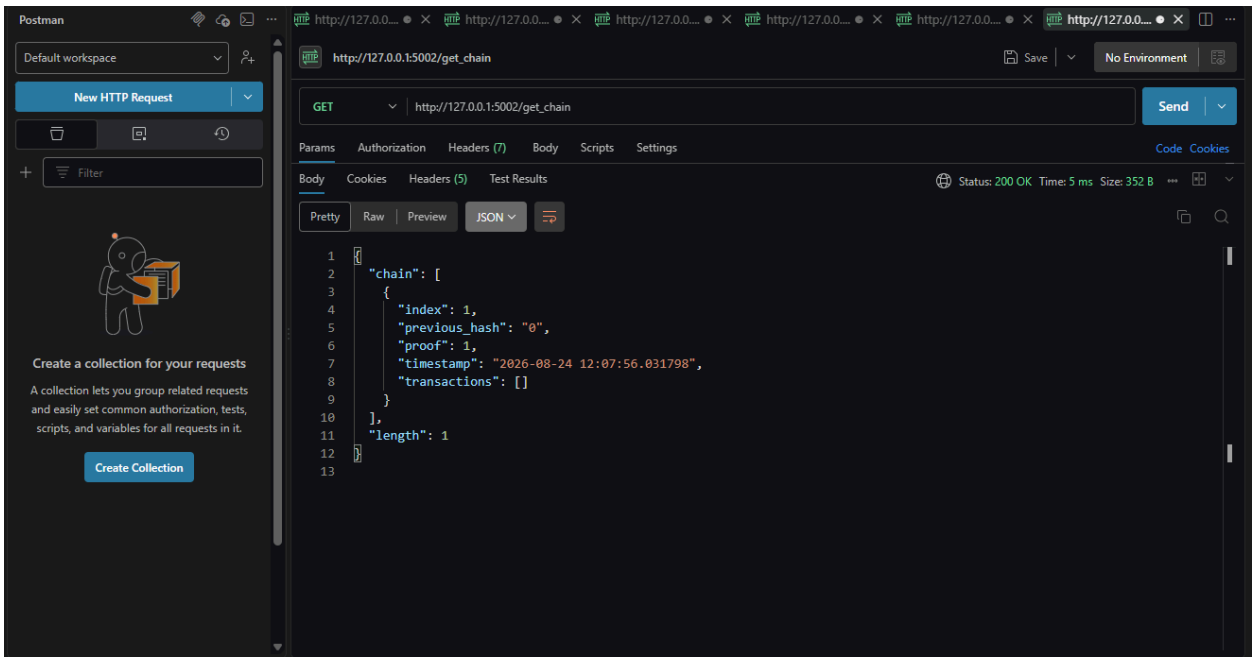
Params Authorization Headers (7) Body Scripts Settings Code Cookies

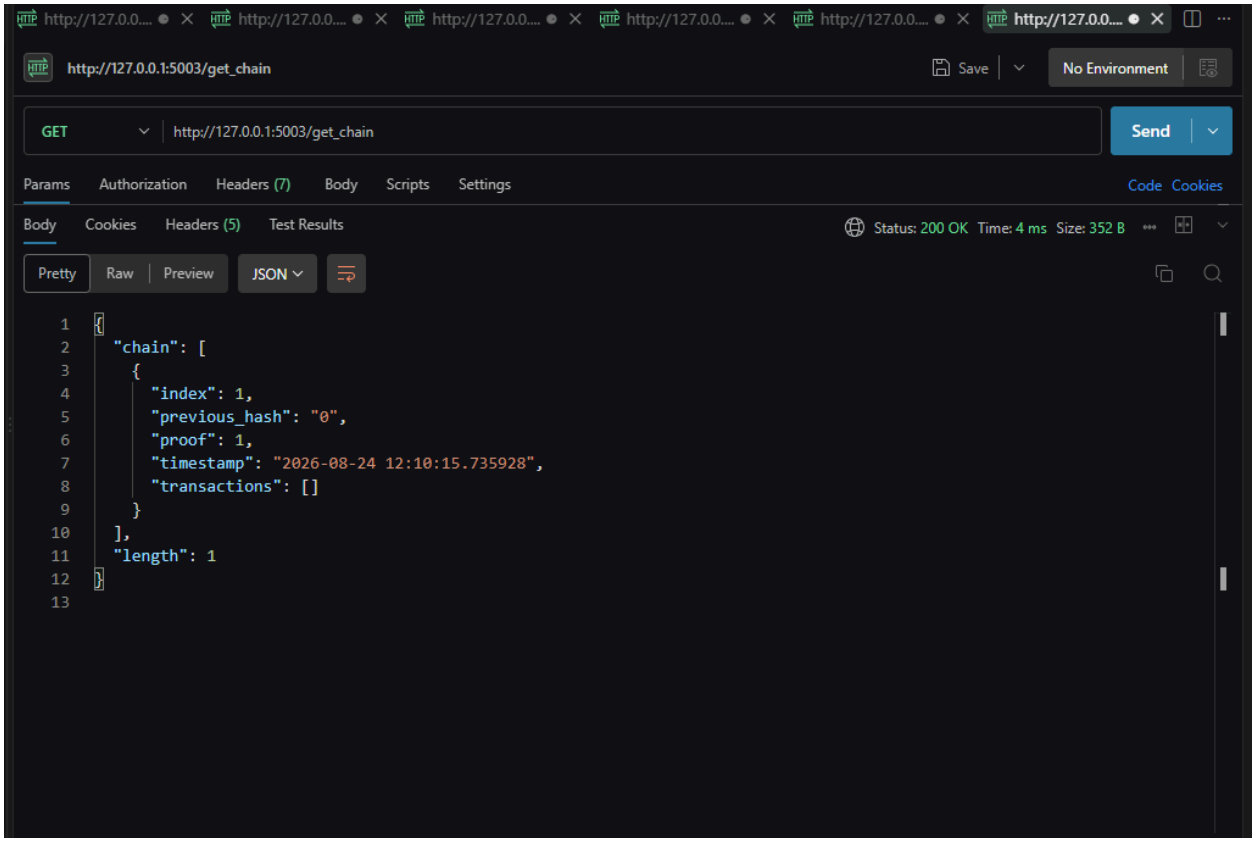
Body Cookies Headers (5) Test Results Status: 200 OK Time: 5 ms Size: 1 KB

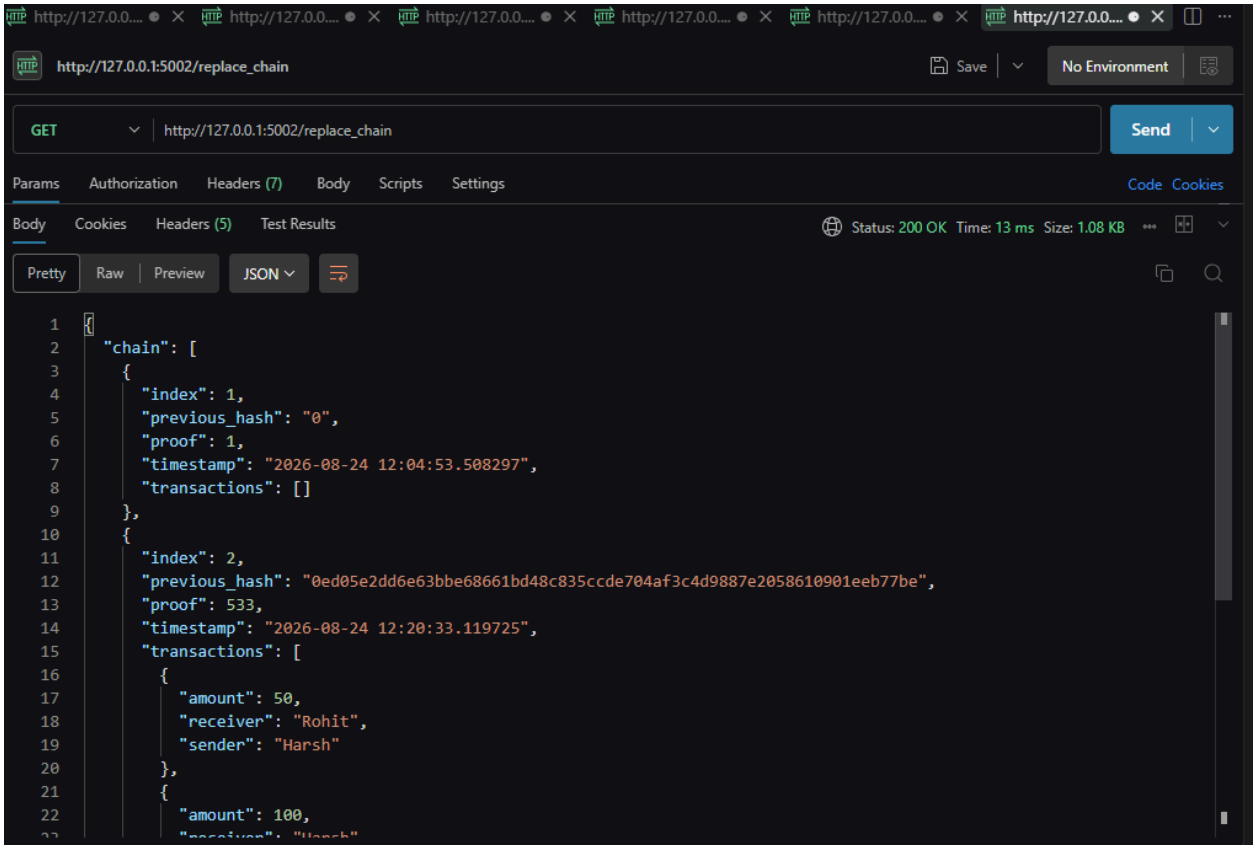
Pretty Raw Preview JSON

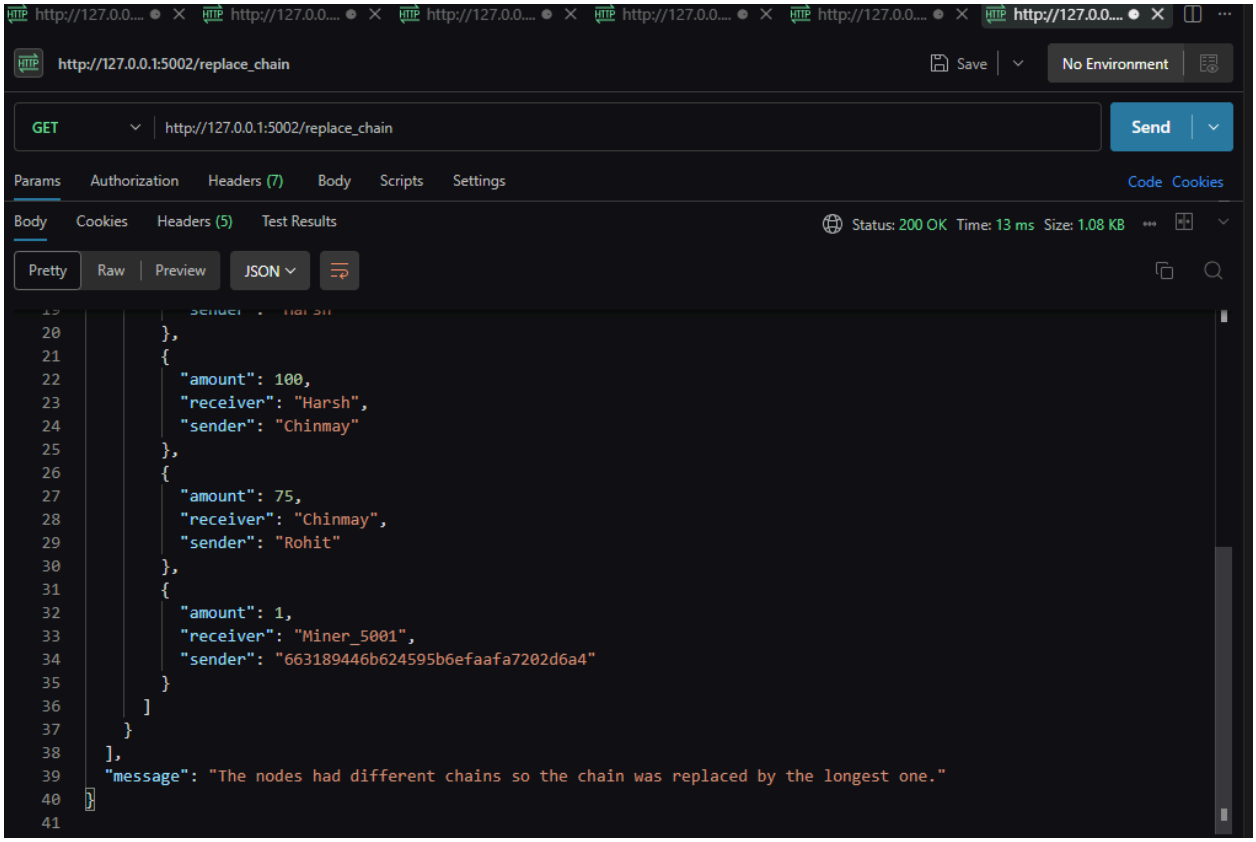

```
{
  "chain": [
    {
      "index": 1,
      "previous_hash": "0",
      "proof": 1,
      "timestamp": "2026-08-24 12:04:53.508297",
      "transactions": []
    },
    {
      "index": 2,
      "previous_hash": "0ed05e2dd6e63bbe68661bd48c835ccde704af3c4d9887e2058610901eeb77be",
      "proof": 533,
      "timestamp": "2026-08-24 12:20:33.119725",
      "transactions": [
        {
          "amount": 50,
          "receiver": "Rohit",
          "sender": "Harsh"
        },
        {
          "amount": 100,
```

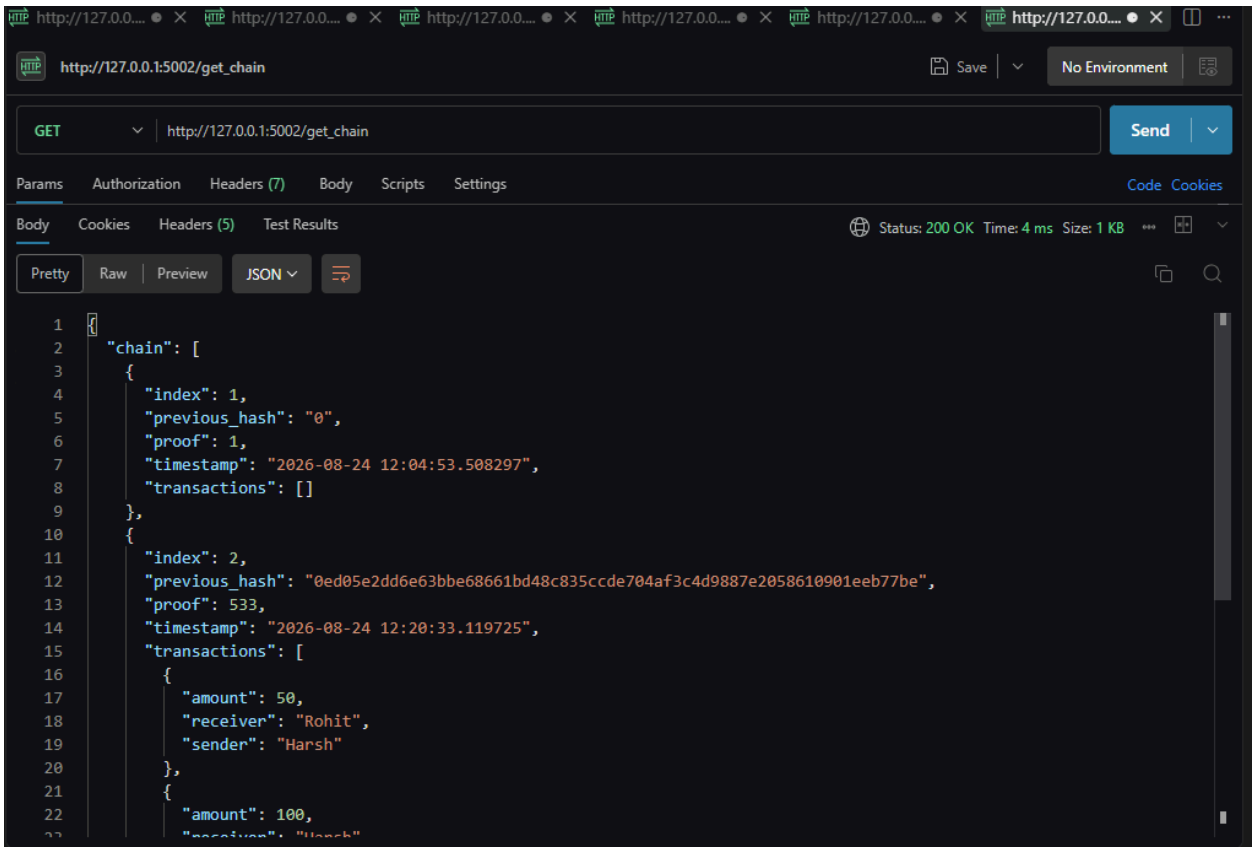

http://127.0.0.1:5001/get_chain
Save
No Environment
GET
http://127.0.0.1:5001/get_chain
Send
Params
Authorization
Headers (7)
Body
Scripts
Settings
Code
Cookies
Body
Cookies
Headers (5)
Test Results
Status: 200 OK
Time: 5 ms
Size: 1 KB
Pretty
Raw
Preview
JSON
{
"amount": 100,
"receiver": "Harsh",
"sender": "Chinmay"
},
{
"amount": 75,
"receiver": "Chinmay",
"sender": "Rohit"
},
{
"amount": 1,
"receiver": "Miner_5001",
"sender": "663189446b624595b6efaafa7202d6a4"
}
]
}
],
"length": 2
}

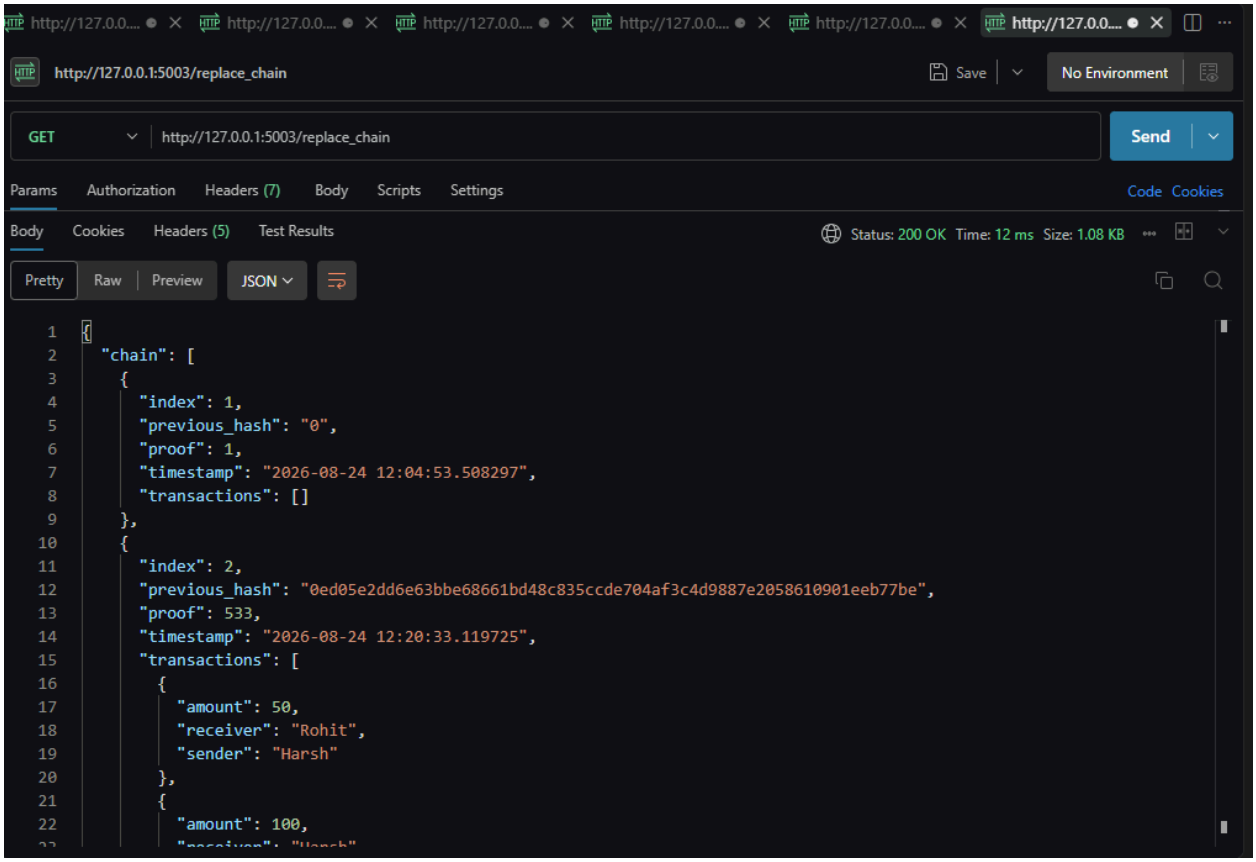

Postman
Default workspace
New HTTP Request
Filter
Create a collection for your requests
A collection lets you group related requests and easily set common authorization, tests, scripts, and variables for all requests in it.
Create Collection
http://127.0.0.1:5002/get_chain
Save
No Environment
GET
http://127.0.0.1:5002/get_chain
Send
Params
Authorization
Headers (7)
Body
Scripts
Settings
Code
Cookies
Body
Cookies
Headers (5)
Test Results
Status: 200 OK
Time: 5 ms
Size: 352 B
Pretty
Raw
Preview
JSON
{
"chain": [
{
"index": 1,
"previous_hash": "0",
"proof": 1,
"timestamp": "2026-08-24 12:07:56.031798",
"transactions": []
}
],
"length": 1
}

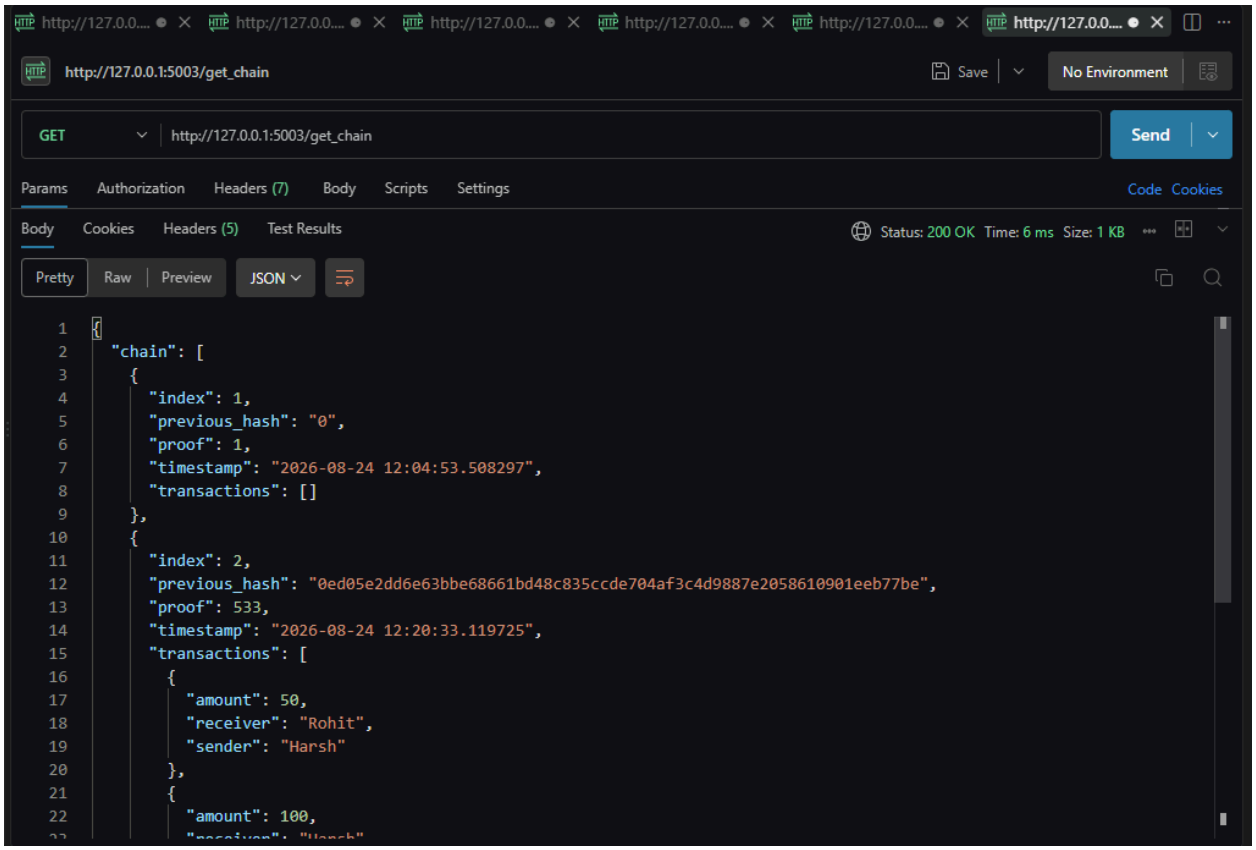

http://127.0.0.1:5003/get_chain
Save
No Environment
GET
http://127.0.0.1:5003/get_chain
Send
Params
Authorization
Headers (7)
Body
Scripts
Settings
Code
Cookies
Body
Cookies
Headers (5)
Test Results
Status: 200 OK
Time: 4 ms
Size: 352 B
Pretty
Raw
Preview
JSON
{
"chain": [
{
"index": 1,
"previous_hash": "0",
"proof": 1,
"timestamp": "2026-08-24 12:10:15.735928",
"transactions": []
}
],
"length": 1
}


http://127.0.0.1:5002/replace_chain
Save
No Environment
GET
http://127.0.0.1:5002/replace_chain
Send
Params
Authorization
Headers (7)
Body
Scripts
Settings
Code
Cookies
Body
Cookies
Headers (5)
Test Results
Status: 200 OK
Time: 13 ms
Size: 1.08 KB
Pretty
Raw
Preview
JSON
{
"chain": [
{
"index": 1,
"previous_hash": "0",
"proof": 1,
"timestamp": "2026-08-24 12:04:53.508297",
"transactions": []
},
{
"index": 2,
"previous_hash": "0ed05e2dd6e63bbe68661bd48c835ccde704af3c4d9887e2058610901eeb77be",
"proof": 533,
"timestamp": "2026-08-24 12:20:33.119725",
"transactions": [
{
"amount": 50,
"receiver": "Rohit",
"sender": "Harsh"
},
{
"amount": 100,


http://127.0.0.1:5002/replace_chain
Save
No Environment
GET
http://127.0.0.1:5002/replace_chain
Send
Params
Authorization
Headers (7)
Body
Scripts
Settings
Code
Cookies
Body
Cookies
Headers (5)
Test Results
Status: 200 OK
Time: 13 ms
Size: 1.08 KB
Pretty
Raw
Preview
JSON
{
"amount": 100,
"receiver": "Harsh",
"sender": "Chinmay"
},
{
"amount": 75,
"receiver": "Chinmay",
"sender": "Rohit"
},
{
"amount": 1,
"receiver": "Miner_5001",
"sender": "663189446b624595b6efaafa7202d6a4"
}
]
}
],
"message": "The nodes had different chains so the chain was replaced by the longest one."
}


http://127.0.0.1:5002/get_chain
Save
No Environment
GET
http://127.0.0.1:5002/get_chain
Send
Params
Authorization
Headers (7)
Body
Scripts
Settings
Code
Cookies
Body
Cookies
Headers (5)
Test Results
Status: 200 OK
Time: 4 ms
Size: 1 KB
Pretty
Raw
Preview
JSON
{
"chain": [
{
"index": 1,
"previous_hash": "0",
"proof": 1,
"timestamp": "2026-08-24 12:04:53.508297",
"transactions": []
},
{
"index": 2,
"previous_hash": "0ed05e2dd6e63bbe68661bd48c835ccde704af3c4d9887e2058610901eeb77be",
"proof": 533,
"timestamp": "2026-08-24 12:20:33.119725",
"transactions": [
{
"amount": 50,
"receiver": "Rohit",
"sender": "Harsh"
},
{
"amount": 100,


http://127.0.0.1:5003/replace_chain
Save
No Environment
GET
http://127.0.0.1:5003/replace_chain
Send
Params
Authorization
Headers (7)
Body
Scripts
Settings
Code
Cookies
Body
Cookies
Headers (5)
Test Results
Status: 200 OK
Time: 12 ms
Size: 1.08 KB
Pretty
Raw
Preview
JSON
{
"chain": [
{
"index": 1,
"previous_hash": "0",
"proof": 1,
"timestamp": "2026-08-24 12:04:53.508297",
"transactions": []
},
{
"index": 2,
"previous_hash": "0ed05e2dd6e63bbe68661bd48c835ccde704af3c4d9887e2058610901eeb77be",
"proof": 533,
"timestamp": "2026-08-24 12:20:33.119725",
"transactions": [
{
"amount": 50,
"receiver": "Rohit",
"sender": "Harsh"
},
{
"amount": 100,


http://127.0.0.1:5003/get_chain
Save
No Environment
GET
http://127.0.0.1:5003/get_chain
Send
Params
Authorization
Headers (7)
Body
Scripts
Settings
Code
Cookies
Body
Cookies
Headers (5)
Test Results
Status: 200 OK
Time: 6 ms
Size: 1 KB
Pretty
Raw
Preview
JSON
{
"chain": [
{
"index": 1,
"previous_hash": "0",
"proof": 1,
"timestamp": "2026-08-24 12:04:53.508297",
"transactions": []
},
{
"index": 2,
"previous_hash": "0ed05e2dd6e63bbe68661bd48c835ccde704af3c4d9887e2058610901eeb77be",
"proof": 533,
"timestamp": "2026-08-24 12:20:33.119725",
"transactions": [
{
"amount": 50,
"receiver": "Rohit",
"sender": "Harsh"
},
{
"amount": 100,